Healthy teams achieve better results.

AI -Based Employee Wellness Management Platform

Support Employee's Mental Health and Sentiment Remotely

Presented By:

Team 4

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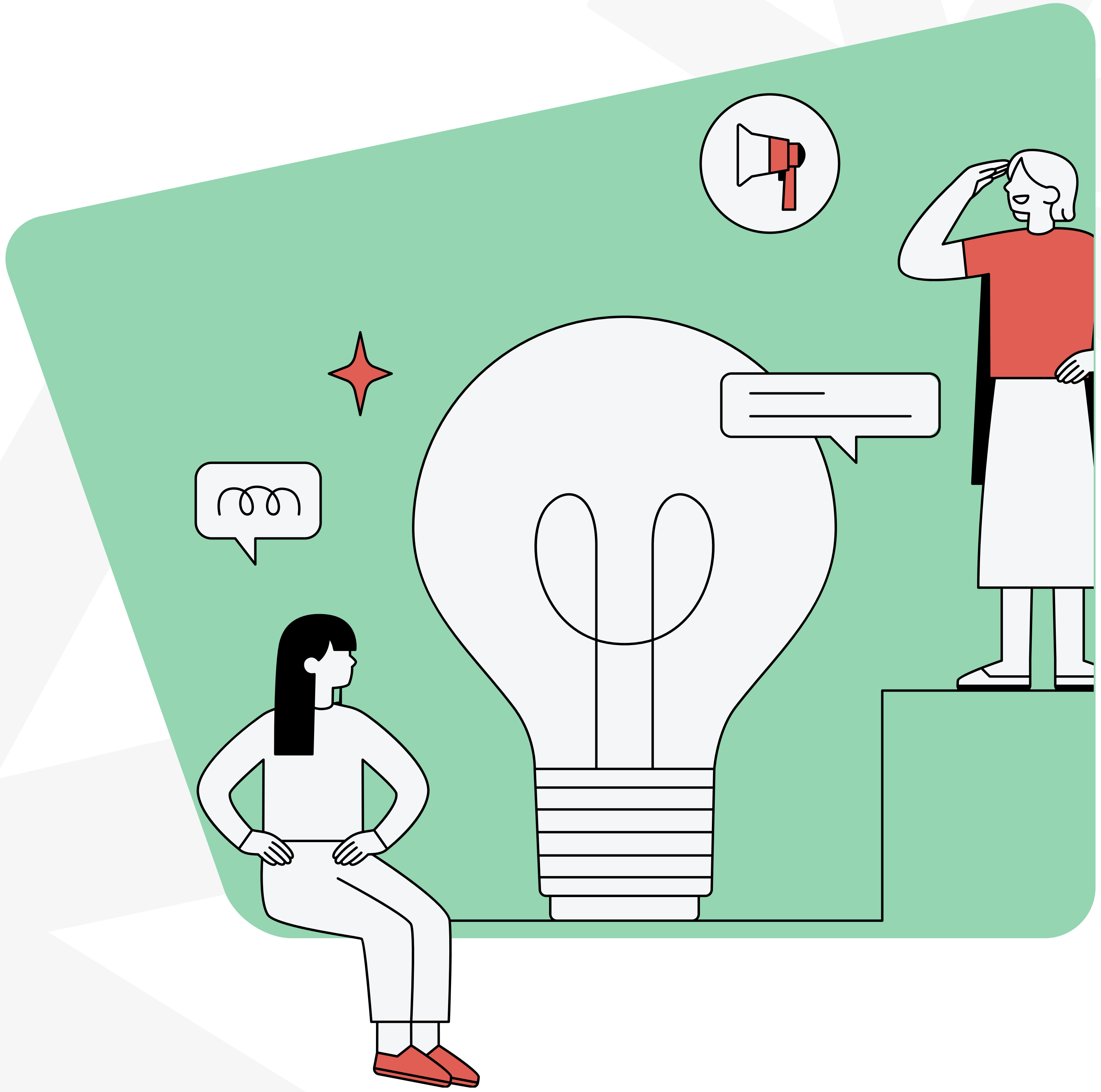
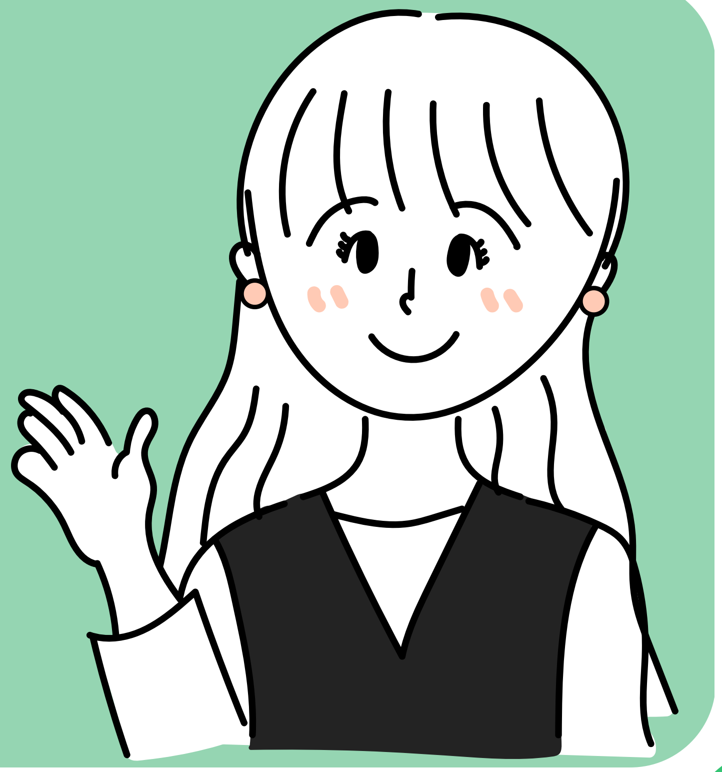


Problem Statement

Current System’s Limitations and Issues

- Lack of a secure and centralized login and wellness management system.
- Difficulty in managing user/admin access and employee wellness records.
- Limited real-time analysis of workforce performance and health trends.
- Need for instant wellness guidance through an intelligent chatbot with natural language response.



Introduction

Knowing about the Project

- Employee wellness is essential for strong productivity and steady growth. The system helps manage health data and spot risks like stress while offering personal wellness tips.
- Employee wellness plays a big role in performance and growth. This system handles health data, predicts risks, and gives each person wellness ideas that support their wellbeing.
- Supporting employee wellness leads to better focus and stronger results. The system works with wellness data, identifies stress risks, and shares recommendations and the sentiment to help everyone stay healthy.

Project Objective

Objectives of the Project

1. Personalized Employee Dashboards: Tracks vitals, sets goals, and delivers custom AI insights.
2. Proactive Admin Analytics: Enables predictive risk modeling, sentiment tracking, and key KPIs.
3. 24/7 AI Wellness Assistant: Provides instant, contextual advice on diet, stress, and exercise.
4. Secure & Scalable Infrastructure: Ensures data privacy, role-based access, and centralized analytics.
5. Automated Admin & Emergency Response: Streamlines claims, scheduling, and instant SOS alerts.

System Architecture

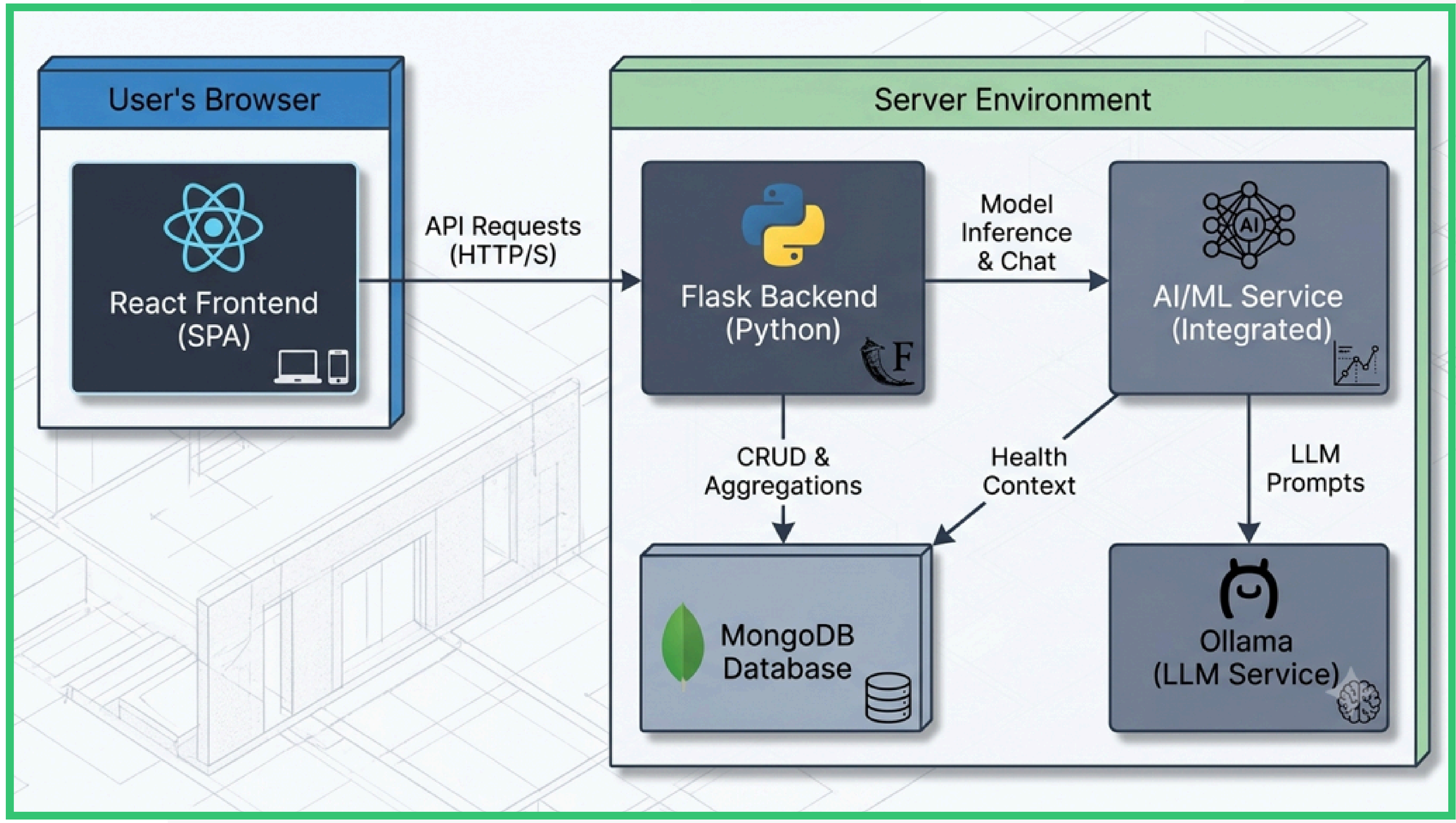
Explaining the structure and flow of the system.

- The application is built on a modern, decoupled architecture, commonly known as a Backend for Frontend (BFF) pattern.

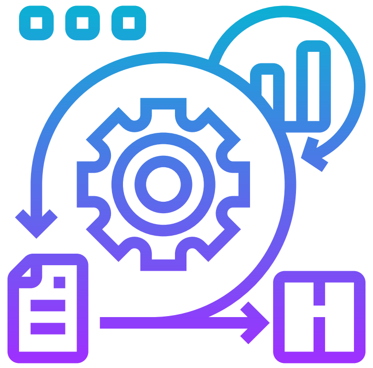
- This means the React frontend is a distinct application that communicates with a dedicated Flask backend via a REST API

- Microservice-Ready AI Integration: Isolates LLM interactions and machine learning model inference into modular background services to maintain high responsiveness and system efficiency.

- Stateless Token Authentication: Utilizes secure JWT authorization across API requests, ensuring strict role-based access control and seamless horizontal scaling.

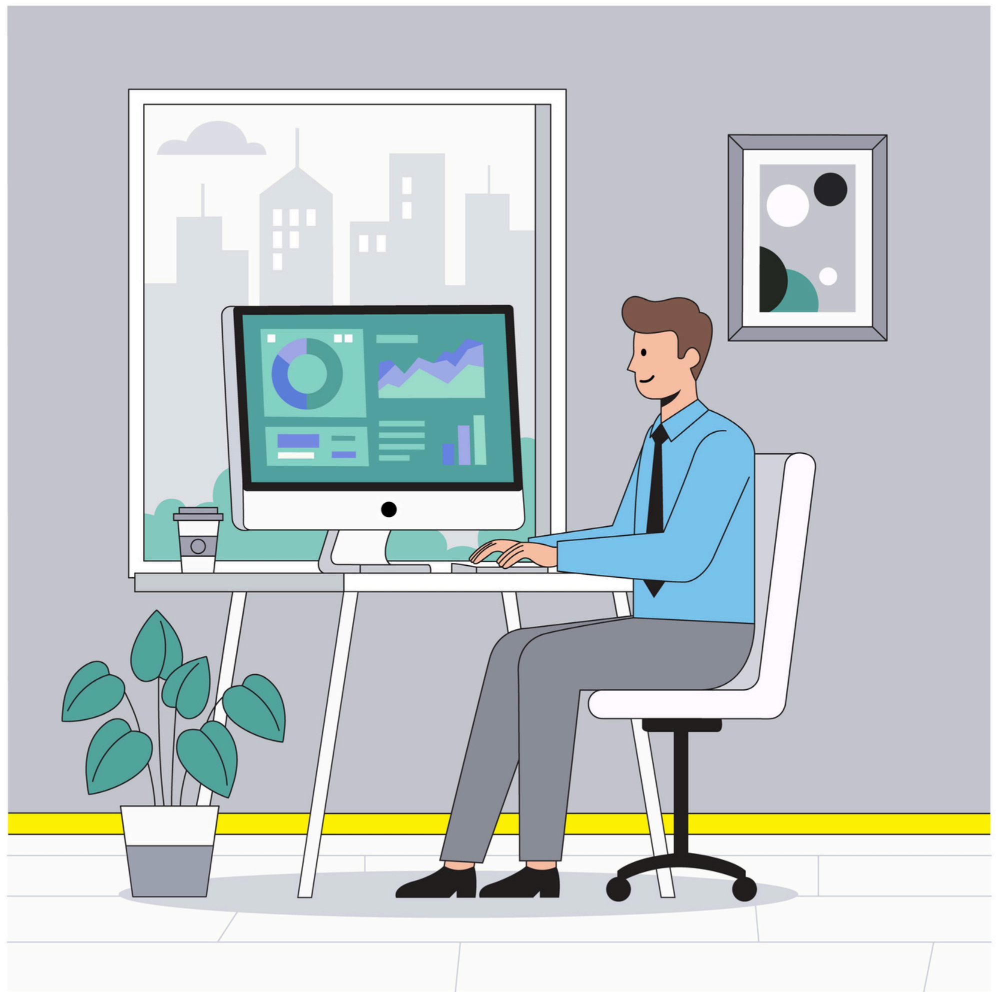


Methodology



How the system was developed and works step-by-step.

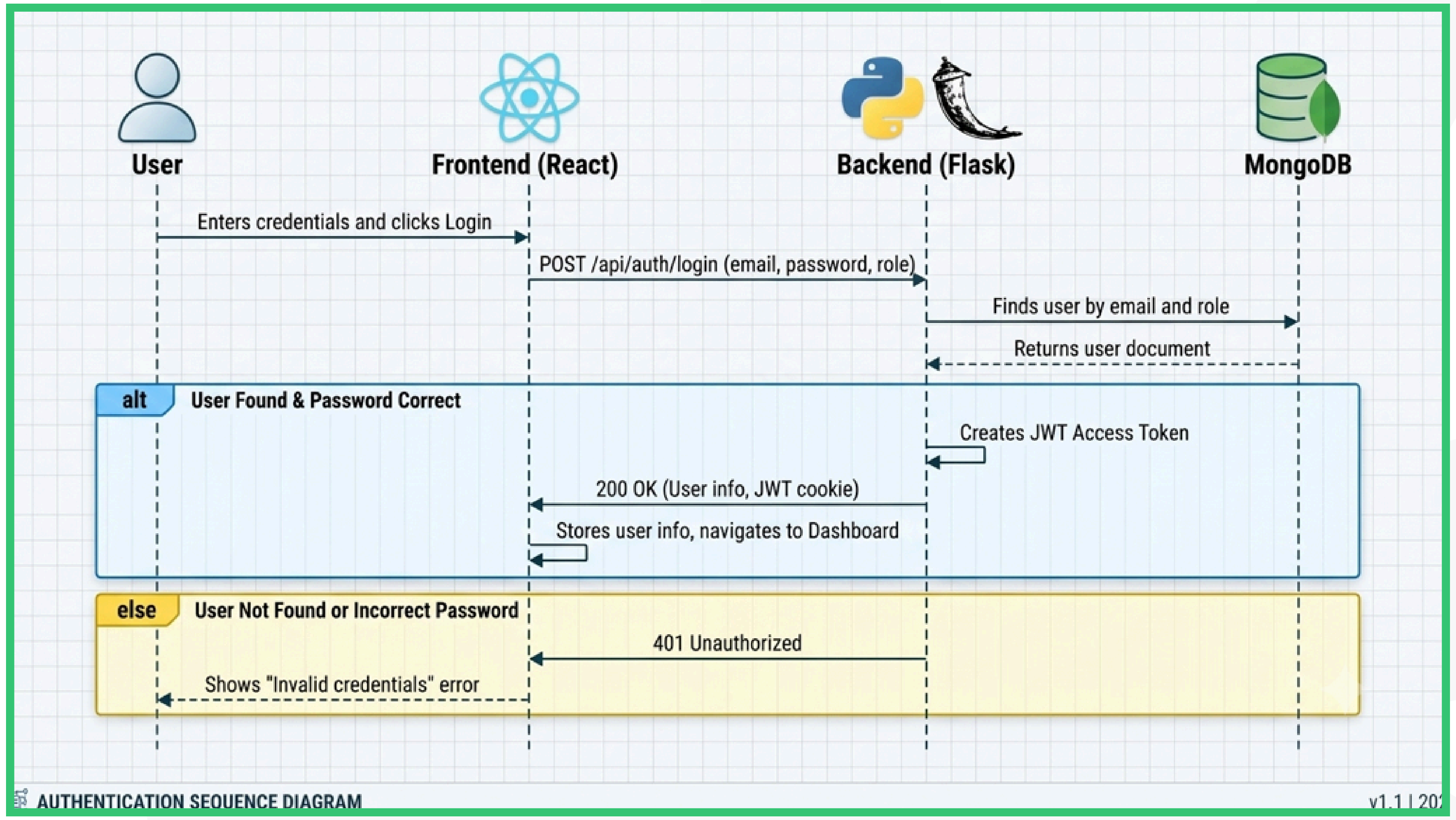
- The system was developed in a modular way, starting with secure user authentication and role-based access for employees and admins.
- After that, the frontend, backend, and database were connected to manage wellness records, analytics, and user interactions smoothly.
- Finally, AI-based features such as risk prediction, recommendations, sentiment analysis, performance analytics, and chatbot support were added to make the platform smarter and more useful.



User Authentication Flow

Employee Journey

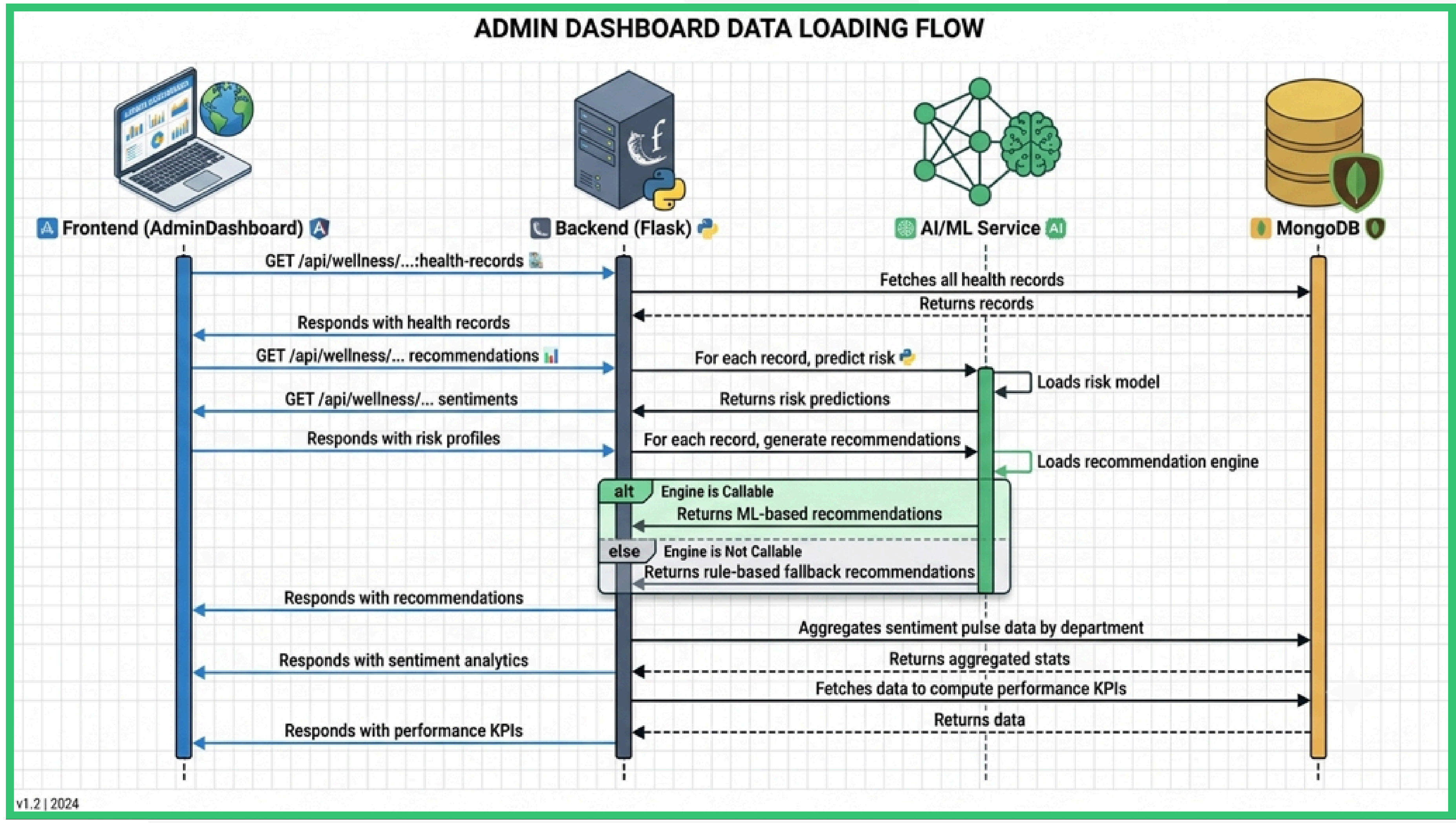
- Auth & Profile: User registers/logs in using JWT authentication → Dashboard loads.
- Data Input: User logs daily habits, sleep, exercise, and mental health feedback.
- Interaction: User requests personalized meal/fitness plans or chats with the InfyWell AI Assistant.
- Action & Emergency: User books checkup appointments or triggers an Emergency SOS alert.



Admin Flow

HR/Management Overview

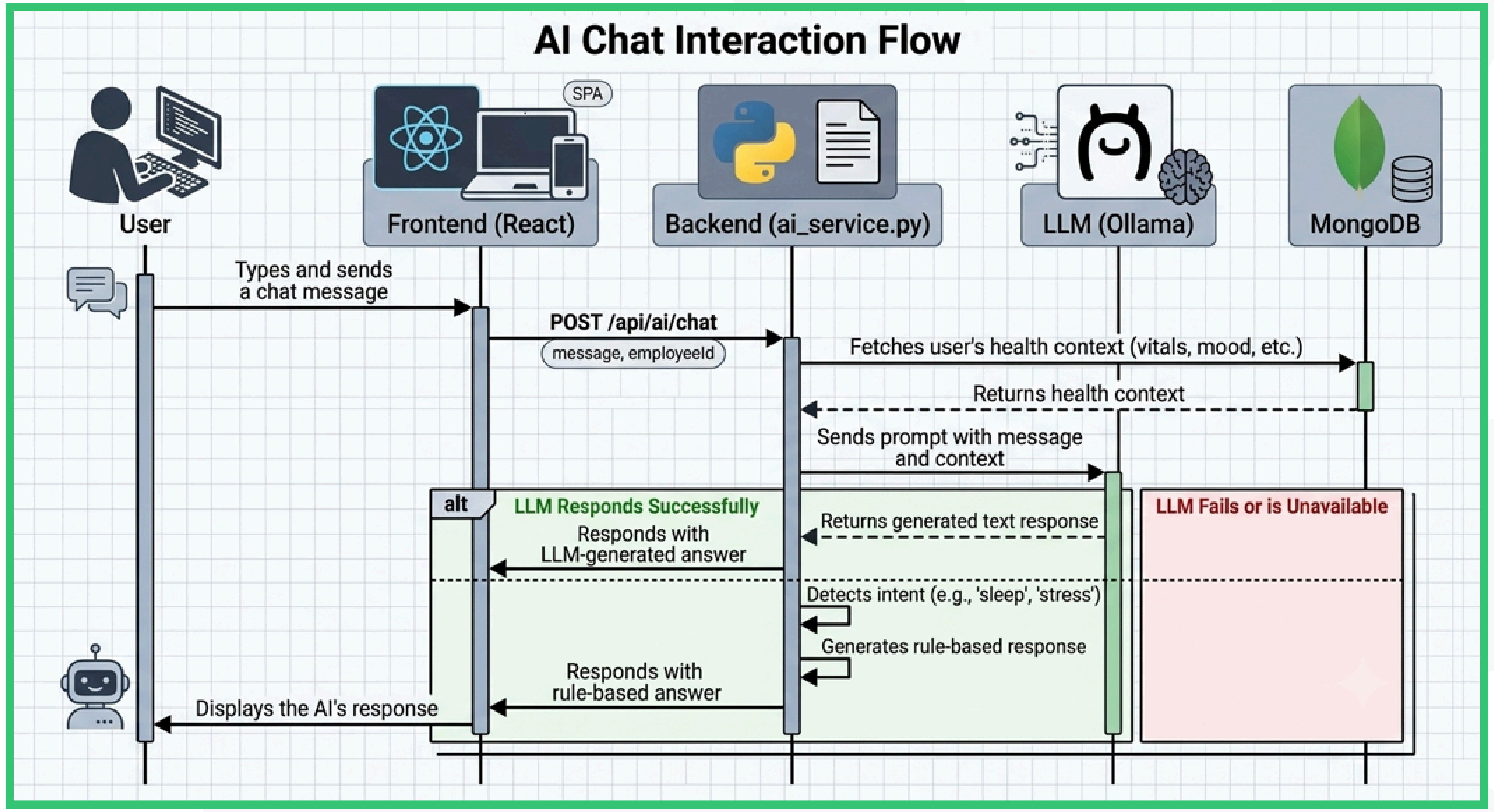
- Dashboard Access: Admin authenticates → Views organization-wide health records and department filters.
- Analytics Monitoring: Admin monitors burnout risk trends, absenteeism rates (3.6%), and overall workforce risk metrics.
- System Management: Admin adjusts risk thresholds (e.g., High Risk = 70), configures privacy settings, manages insurance claims, and sends broadcast notifications.
- Emergency Handling: Admin receives real-time SOS alerts and health checkup logs.



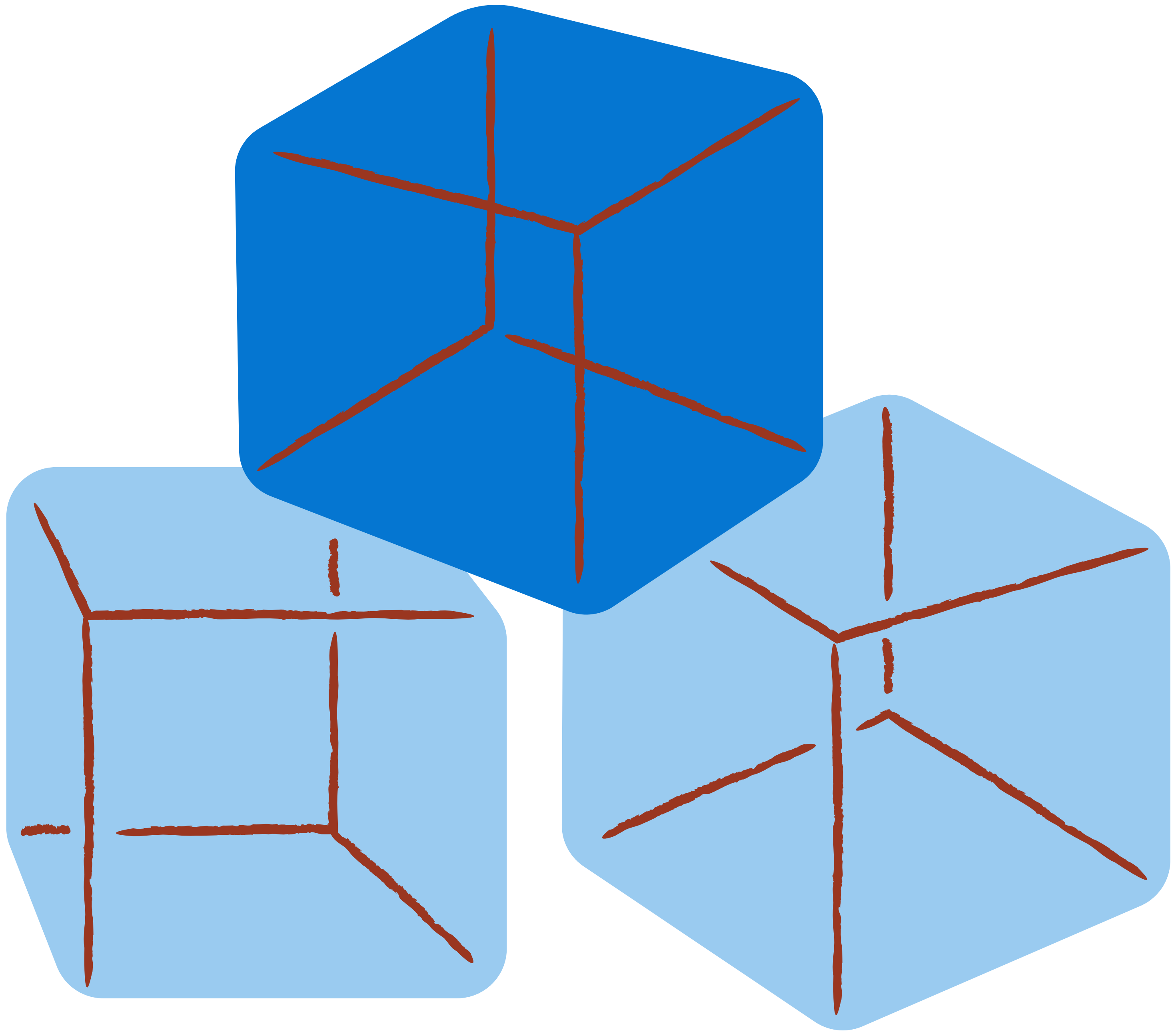
AI Chat Interaction Flow

Data Processing Engine

- Data Ingestion: Collects raw health metrics, BMI, blood pressure, and text feedback from MongoDB.
- Inference Pipeline:
- ML Models (Random Forest/Decision Tree): Calculates risk severity scores (<45% Low, 45-60% Moderate, >75% High).
- NLP Engine: Performs sentiment and emotion analysis on employee feedback/logs.
- Ollama LLM Service: Ingests health context and user prompts to generate personalized daily tips, diets, and conversational advice.
- Output Delivery: Renders predicted risks and recommendations back to React Frontend dashboards in real time.



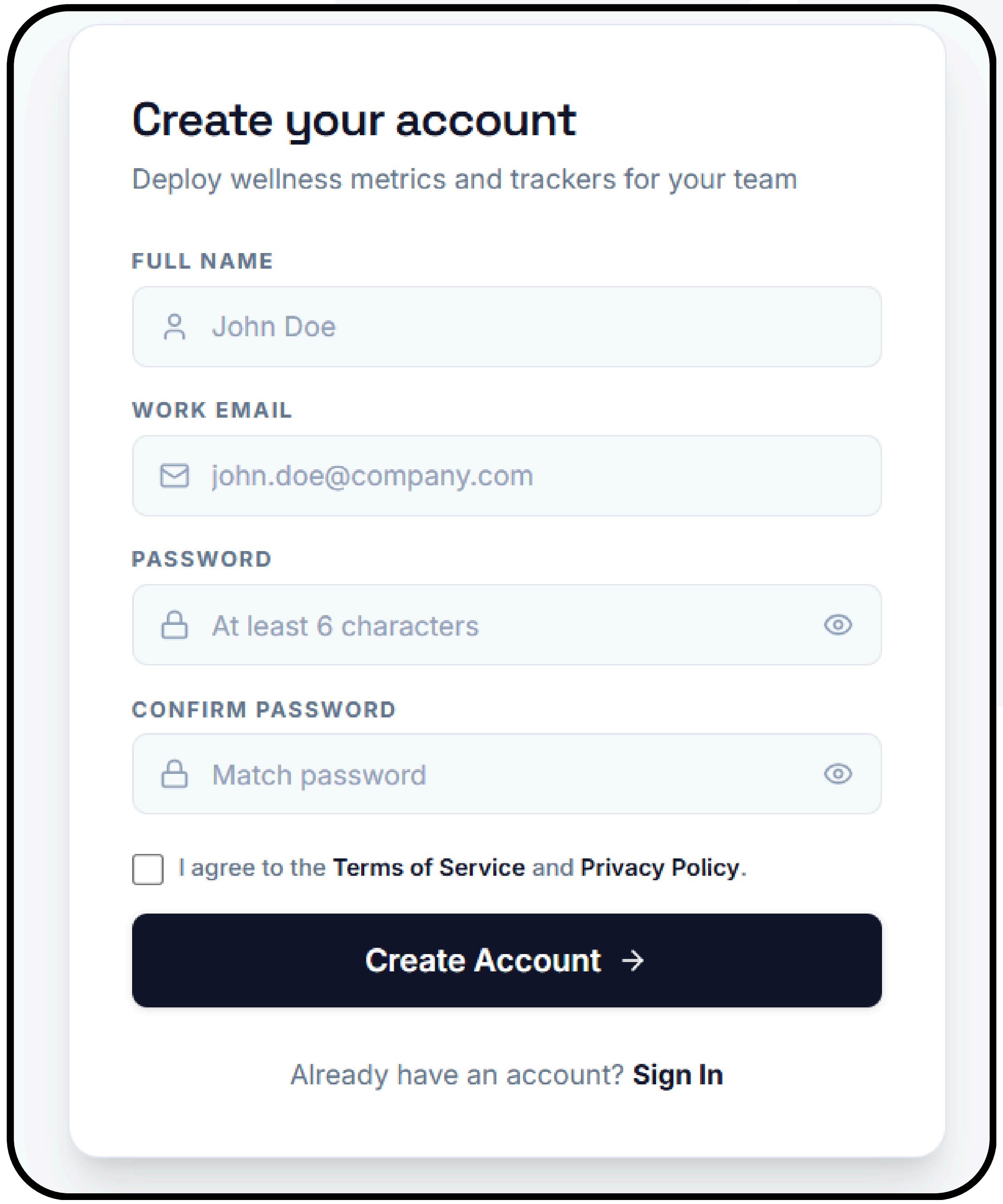
Core Modules

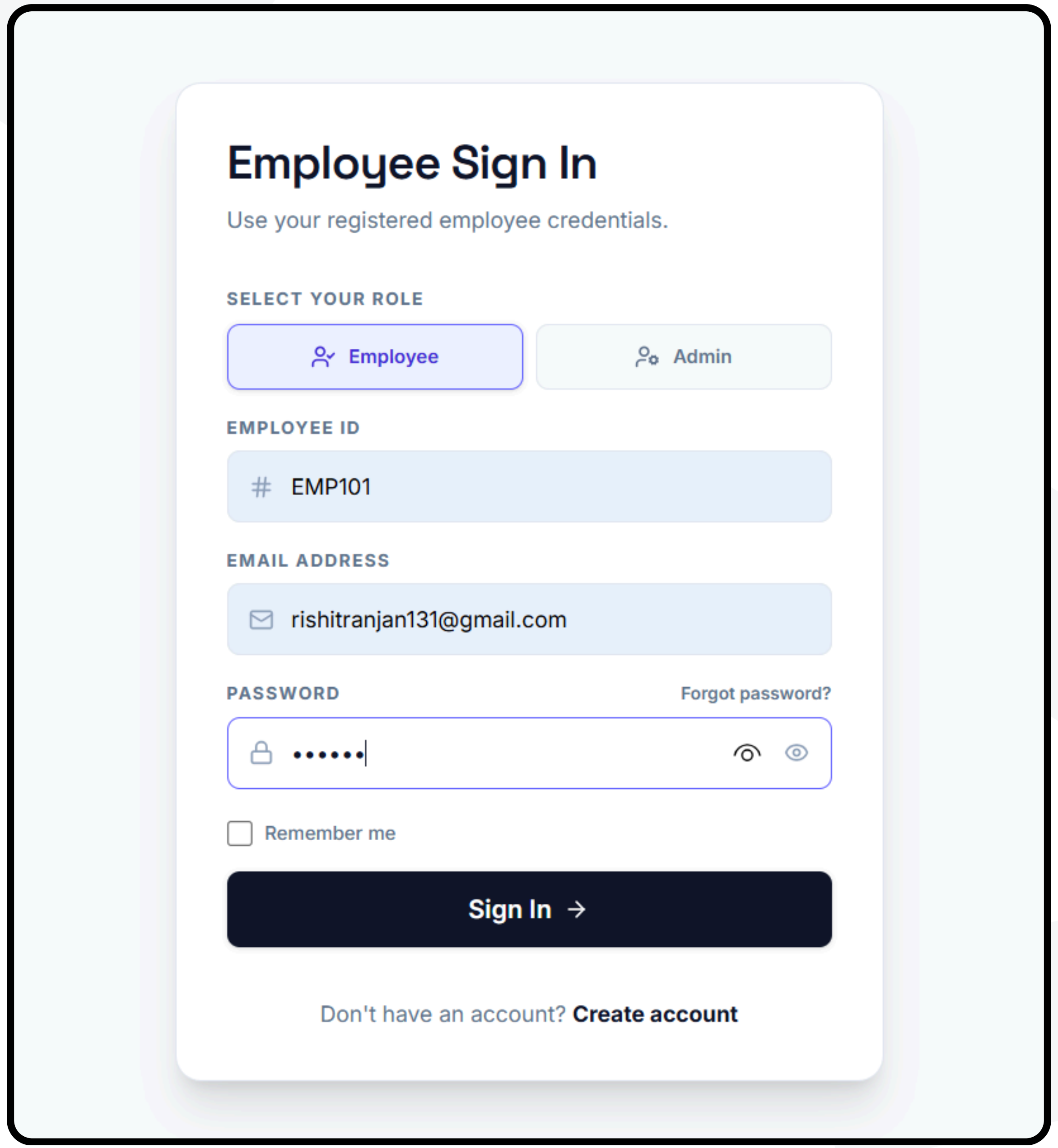


Login & SignUp Module

SignUp and SignIn

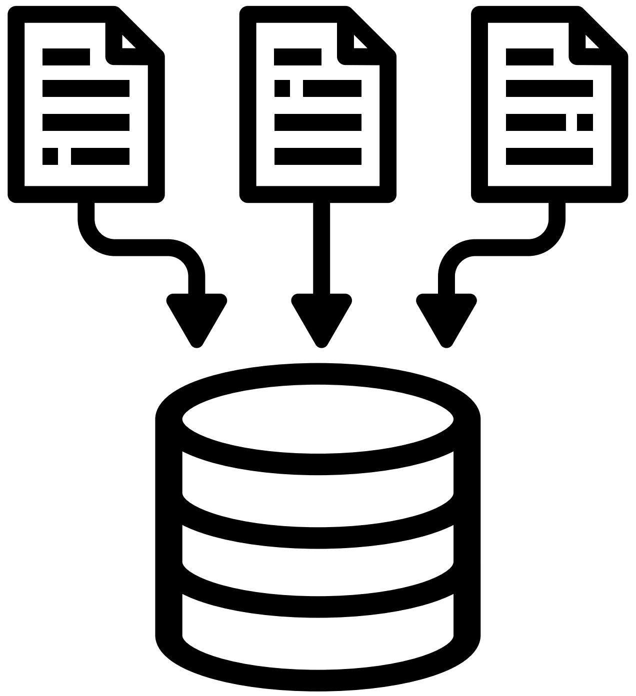
- User login with email and password.
- User registration with required fields.
- Remember me option in login form.
- Forgot password flow with email, OTP, and reset password steps in the frontend prototype.
- Role-based redirect to user or admin dashboard after authentication.





LOGIN!

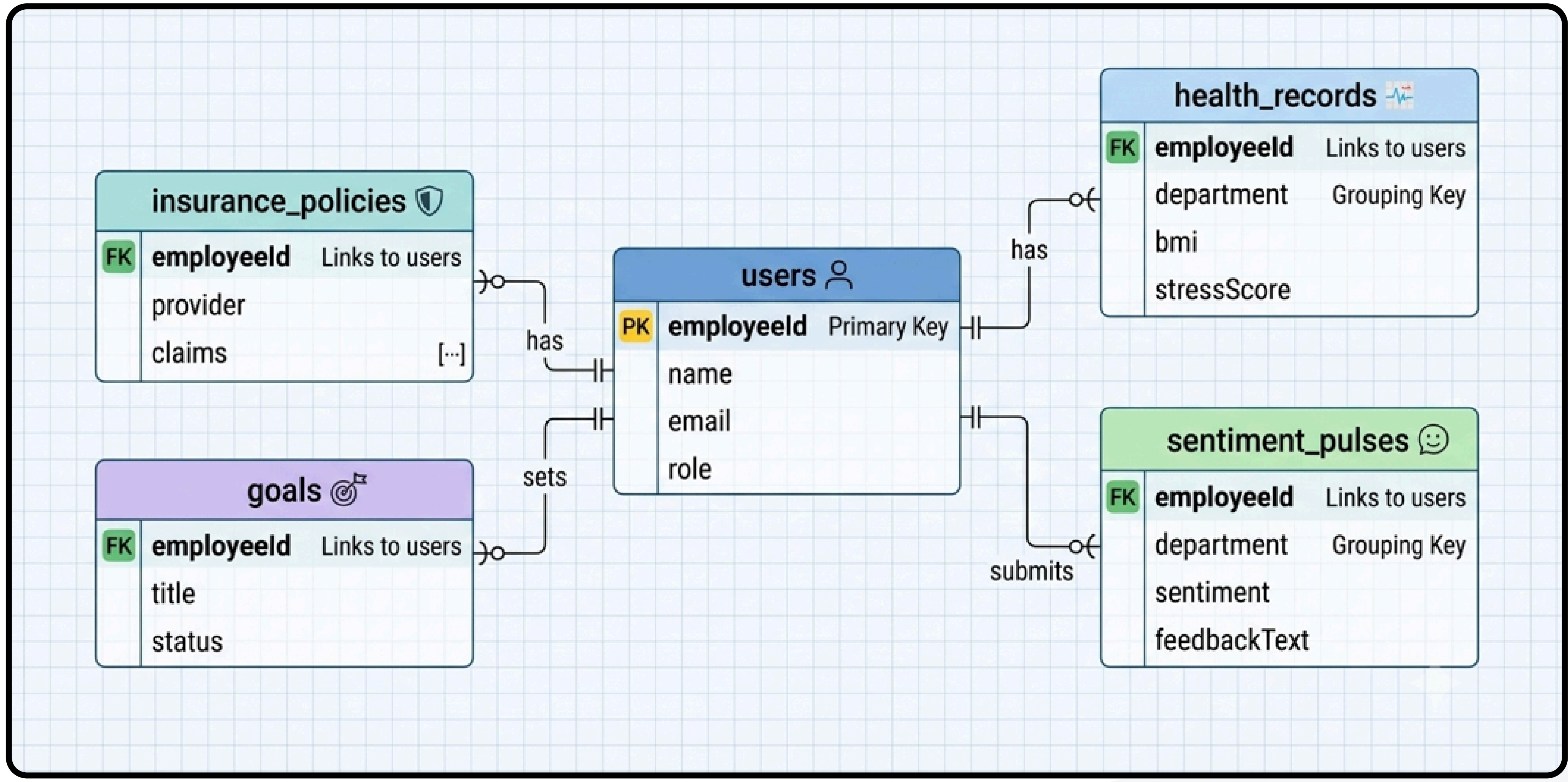
Database



Database Management of the Employee Data

The backend connects to MongoDB using PyMongo.

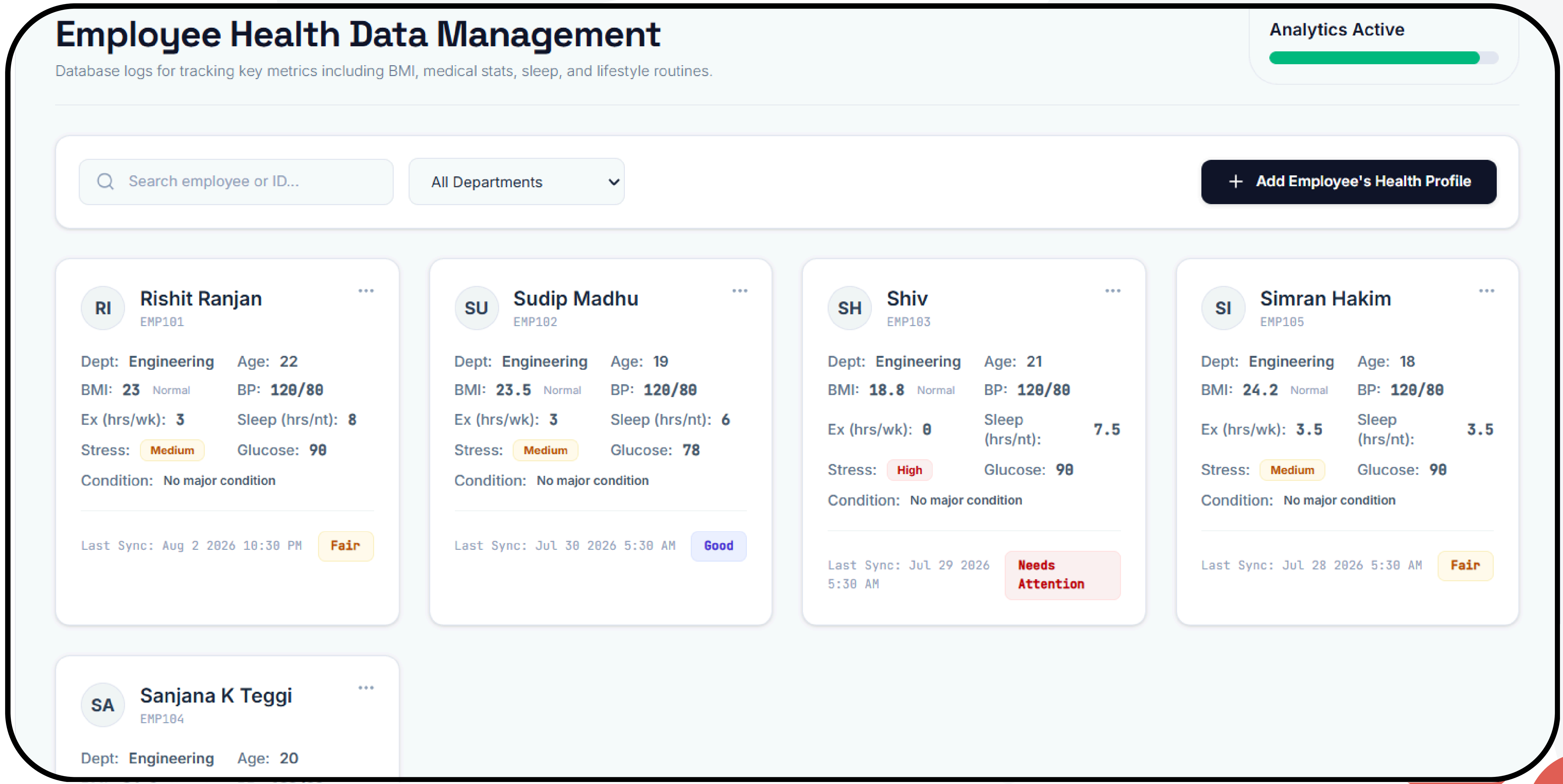
User-related data is stored in collections including admin and employees in JSON - like format.



Health Records Collection

Collecting the Health Record

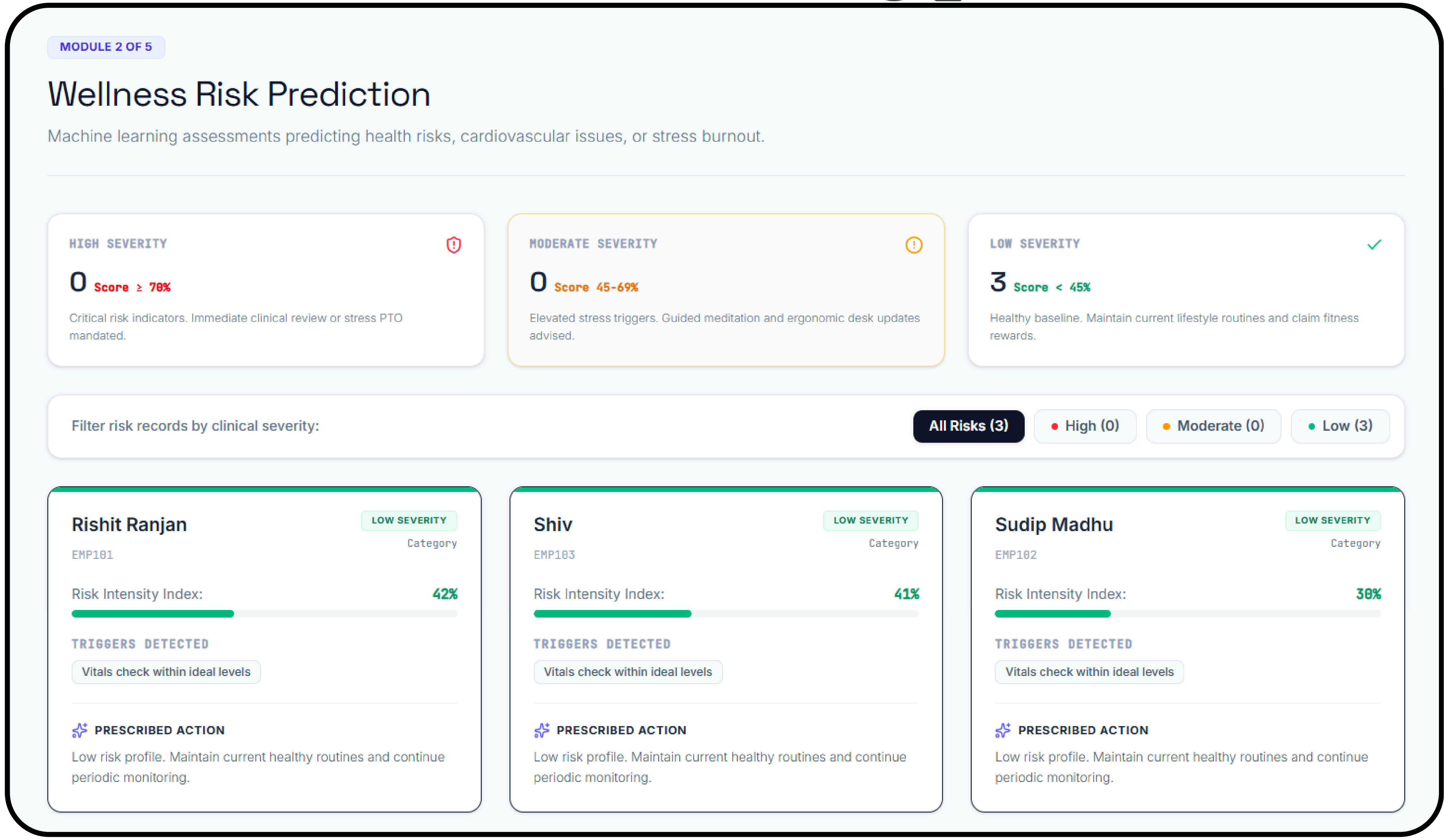
- BMI and medical reports.
- Exercise and sleep tracking.
- Attendance and assessments.
- Data preprocessing and validation.



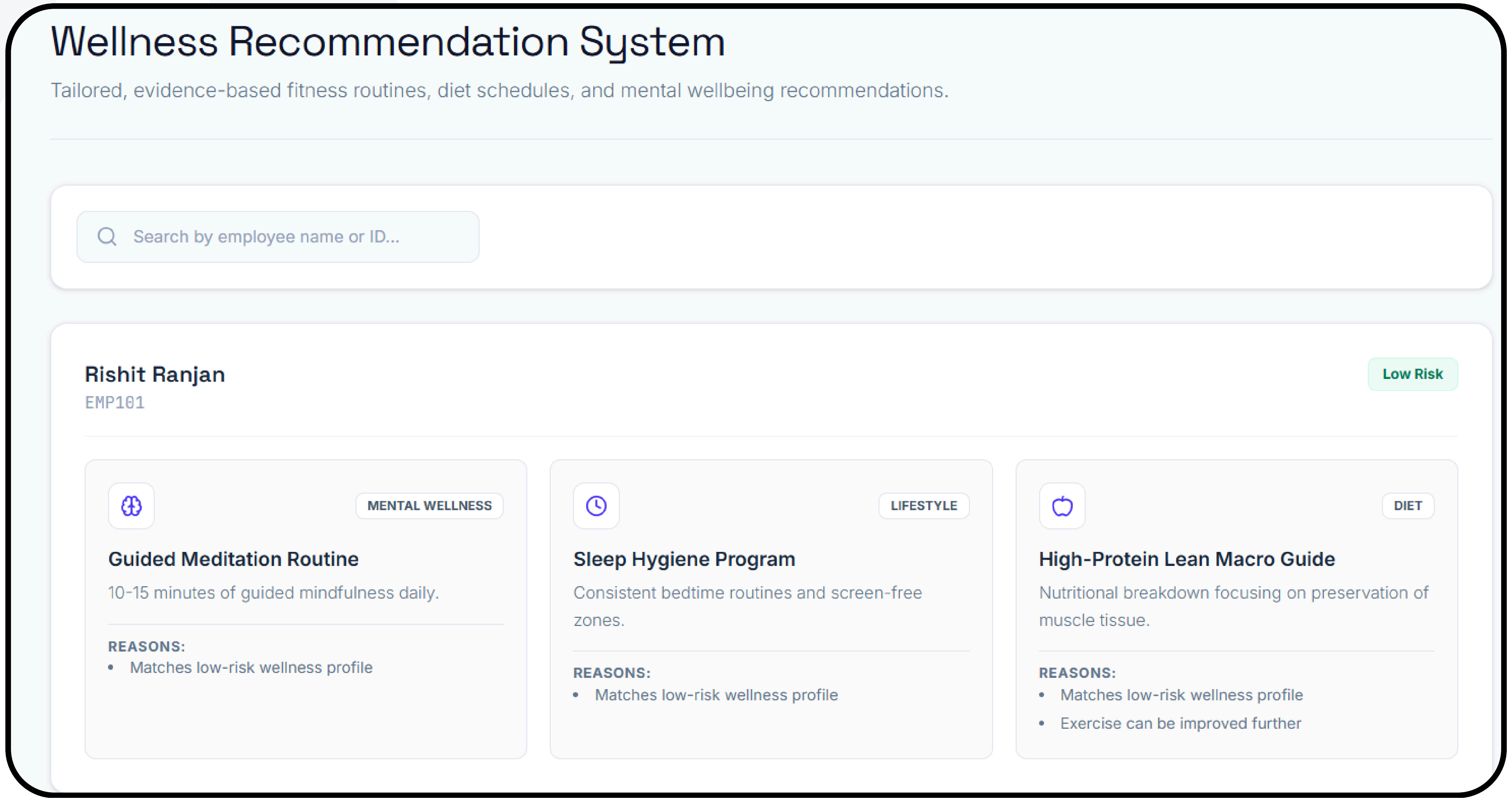
Wellness Risk Prediction

Predicting the Risk

- Predicts stress and burnout.
- Detects obesity and hypertension risk.
- Uses historical and real-time data.
- ML: Decision Tree, Random Forest



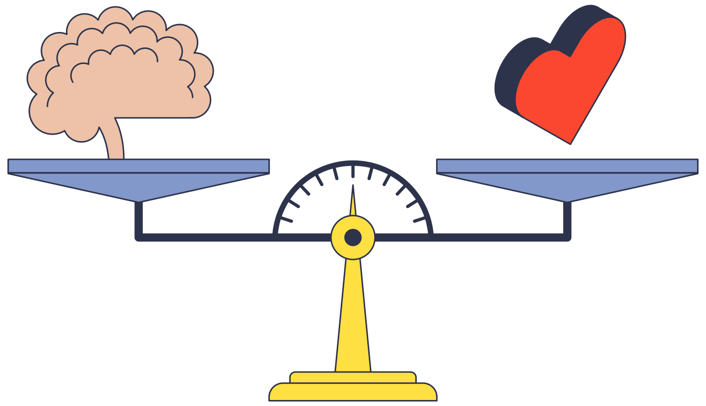
Personalized Wellness Recommendation

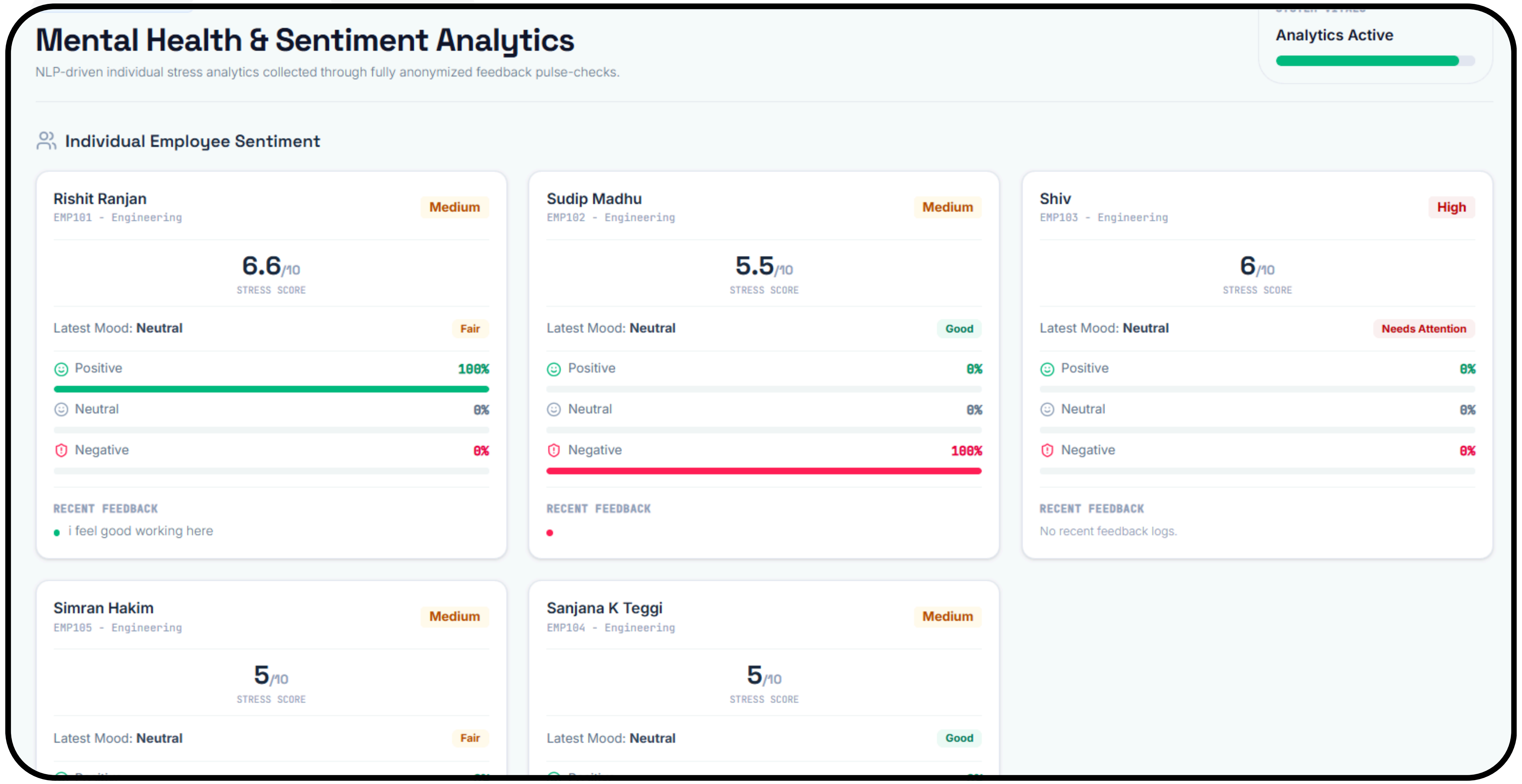


Employee Wellness Recommendation

- Personalized fitness plans
- Diet and lifestyle suggestions
- Mental wellness and yoga activities
- AI: collaborative filtering, content-based filtering, Generative AI

Mental Health & Sentiment Analysis

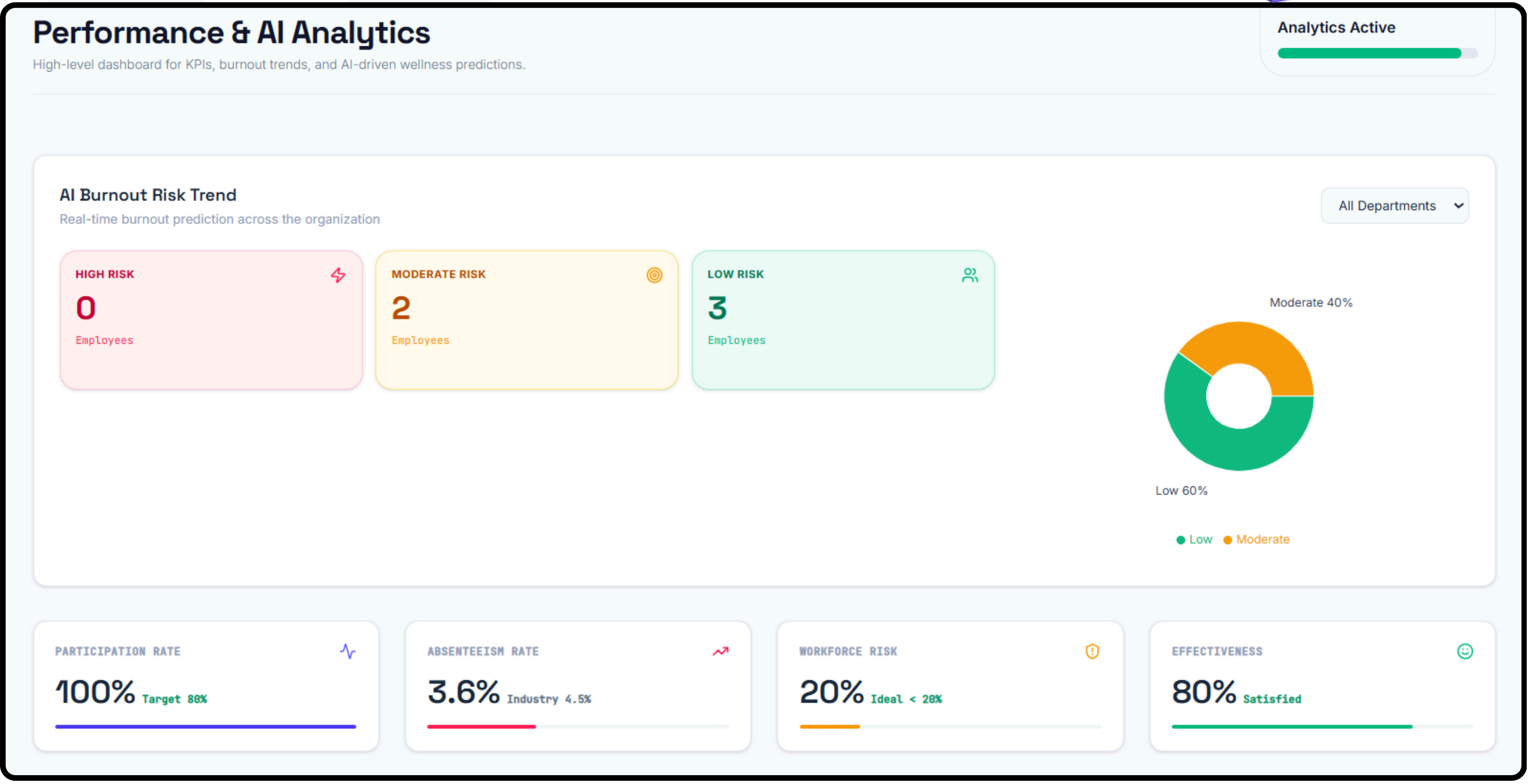




Sentiment and Mental Health

- Analyzes surveys and feedback.
- Can use emails with consent and chatbot conversations
- Detects stress, anxiety, burnout.
- NLP, sentiment analysis, emotion detection

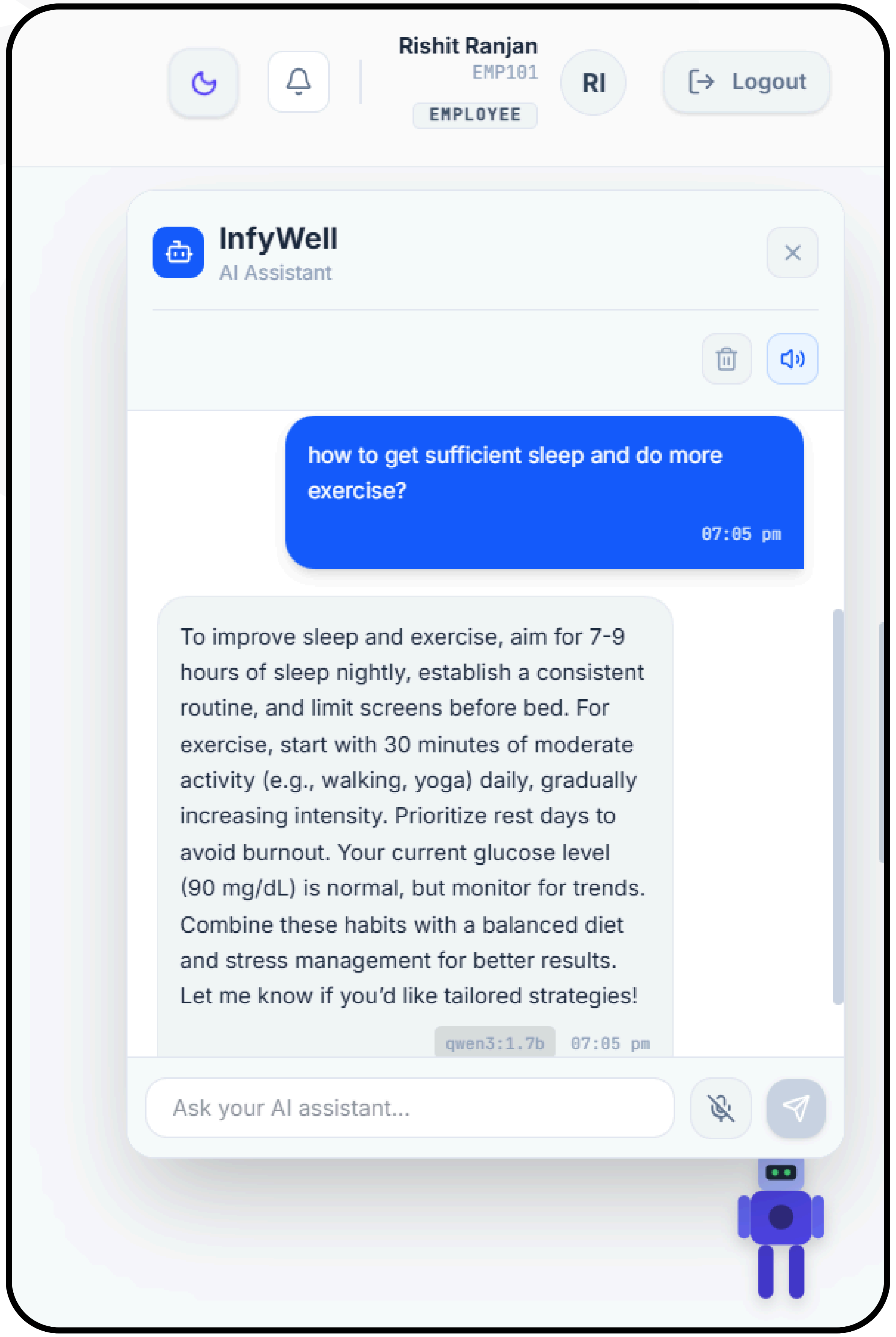
Performance & AI Analysis

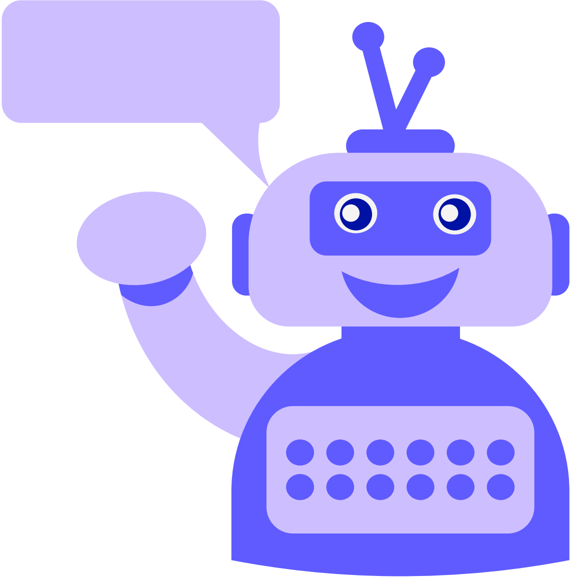


Employee Performance Metrics

- Tracks absenteeism and workforce health risk.
- Measures wellness program effectiveness.
- Helps admin monitor employee performance trends.

Chatbot Module





Chatbot Support and Assistance

- AI-powered wellness assistant for employees.
- Answers questions on sleep, diet, exercise, stress, and burnout.
- Provides interactive chat support inside the user dashboard.

Additional Features

Admin Dashboard

- Insurance Management
- Notification Center
- Health Checkups
- Emergency SOS
- Health Expenses
- System Settings

Employee Dashboard

AI Wellness Assistant
AI Meals and Diet Plans
Health Reports and Timeline
Health Checkups
Emergency SOS trigger alert
Health Expenses claims

Conclusion

- The project includes secure authentication, database integration, risk prediction, recommendation, sentimental analysis, performance analytics, and an AI wellness chatbot in one platform.
- It supports both employee and admin roles for managing wellness data and system features.
- These modules make the system more intelligent, interactive, and useful for future expansion.
- Modular Scalability: Designed with an extensible architecture that allows seamless integration of future health-tech tools and wearable devices.
- Real-time Incident Response: Combines continuous health metric tracking with automated SOS triggers and expense management for rapid emergency handling.

Thank you so much for your time.

Project demo. and code